

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	M Rakesh	Reg. No.:	192421006
Section / Batch:	IT	Date:	8/1/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT**Code of Conduct**

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				

Total Marks (100):	
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Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1, An inventory management system searches for product IDs in an unsorted dataset using Linear Search. a) Explain how Linear Search is applied. b) Analyze performance issues. c) Suggest improvements and justify.

Inventory Management System – Linear Search

a) Explain how Linear Search is applied

- Linear Search checks each product ID one by one from the beginning of the list.
- It compares every element with the required product ID.
- If a match is found, it returns the position.
- If no match is found, it returns "Not Found."

b) Analyze performance issues

- Best Case: **$O(1)$** (element found at the first position)
- Average Case: **$O(n)$**
- Worst Case: **$O(n)$** (element is last or not present)
- Slow for large unsorted datasets because every element may need to be checked.

c) Suggest improvements and justify

- Sort the data and use **Binary Search** for faster searching.
- Use a **Hash Table** for nearly constant-time searching.
- These methods reduce search time and improve performance for large datasets.

2, A payroll system uses Binary Search on sorted employee IDs, but frequent insertions disturb order. a) Explain Binary Search operation. b) Analyze the impact of unsorted data. c) Discuss maintaining sorted data for efficiency.

a) Explain Binary Search operation

- Binary Search works only on **sorted data**.
- It compares the target employee ID with the middle element.
- If equal, the search stops.
- If smaller, search the left half.
- If larger, search the right half.
- This process repeats until the element is found or the search space becomes empty.

b) Analyze the impact of unsorted data

- Binary Search does **not work correctly** on unsorted data.
- Frequent insertions can disturb the sorted order.
- The algorithm may return incorrect results or fail to find an existing employee ID.

c) Discuss maintaining sorted data for efficiency

- Insert new employee IDs in their correct sorted position.
- Re-sort the data after multiple insertions if necessary.
- Use balanced data structures (e.g., BST or AVL Tree) to maintain sorted order.
- Keeping the data sorted ensures Binary Search continues to run in **$O(\log n)$** time.

Time Complexity (Binary Search):

- Best Case: **$O(1)$**
- Average Case: **$O(\log n)$**
- Worst Case: **$O(\log n)$**
- Space Complexity: **$O(1)$** (iterative implementation)